# MIT805 Project: PART 1 Data Collection and Analysis

Group 3

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## 1 Dataset Provenance and Description

The dataset used for this project is the *Yellow Taxi Trip Data* published by the New York City Taxi and Limousine Commission (TLC) through NYC Open Data. Each record represents a single taxi trip and contains information such as pickup and drop-off times, passenger counts, trip distance, payment method, and fare amounts. The dataset is publicly available [1] and is commonly used for academic research and transportation analytics [2]. GitHub Repository:

`https://github.com/Siya1314/MIT-805-Project`

## 2 Dataset Size and Characteristics

| Characteristic | Description |
|---|---|
| Source | NYC Open Data |
| Provider | NYC Taxi & Limousine Commission |
| Format | CSV / Parquet |
| Last Updated | May 2026 |
| Attributes | 20 Columns |

Table 1: Dataset Characteristics

The raw dataset occupies approximately 27 GB of storage and consists of monthly Yellow Taxi trip records stored in Parquet format, covering the period from January 2009 to May 2026. The working dataset size is approximately 13 GB and consists of records from January 2011 to December 2017. The processed dataset size is approximately 3.2 GB and contains records from January 2016 to December 2017. Due to the large volume of data, data processing and transformation were performed using PySpark, while data analysis and visualisations were produced using Pandas and Matplotlib.

## 3 Data Quality and Preparation

The following preprocessing steps were performed:

- Only looked at rides where the passenger counts were greater than 0.
- Converted timestamps to datetime format.
- Removed invalid trip distances.
- Only looked at fare amounts greater than 0.
- Created variables such as trip duration, time of day, and day of week.

Several data quality concerns were identified:

- Missing location information in some records.
- Potential inaccuracies in driver-reported fields such as high passenger counts.
- Dataset only represents yellow taxi trips.
- Data reflect transportation patterns only from 2016 to 2017.

## 4 Exploratory Data Analysis

The exploratory analysis revealed several interesting patterns:

- Most trips were shorter than 5 miles.
- Demand peaked during the evening, with the largest peak on Saturdays.
- Credit card payments were more common than cash payments.
- Most trips happened during the months of March, April and May.
- Outliers have been identified, such as drivers reporting to have more than 6 passengers, fare amounts of more than $200 and trips of more than 50 miles.

Figures generated during the EDA process are included in Appendix A.

## 5 Big Data Analysis Using the 7 Vs

1. **Volume**: Over 240 million records require scalable storage and processing.
2. **Velocity**: Taxi trips are generated continuously throughout the day, enabling near real-time analytics.
3. **Variety**: The dataset contains temporal, numerical, categorical, and geographic data.
4. **Value**: Data-driven insights can improve transportation planning and fleet management.
5. **Veracity**: Data quality issues such as missing values and reporting errors affect reliability.
6. **Variability**: Demand fluctuates across seasons, and weekdays.
7. **Visualization**: Trend charts help stakeholders understand travel patterns.

## 6 Business and Societal Value

The dataset provides significant value for both business and public-sector decision making.
**Business Value**

- Dataset can be used for demand forecasting.
- Revenue optimization.
- Fleet allocation and route planning.

**Societal Value**

- It can be used to assist with urban transport planning.
- It can be used to manage traffic congestion
- It can also be used for infrastructure investment decisions.

**Limitations**

1. Only yellow taxi trips are represented.
2. Since only data from 2016 to 2017 has been analysed, the dataset may not reflect current transportation behaviour.

## References

[1] New York City Taxi and Limousine Commission. 2026. *TLC Trip Record Data*. Available at: `https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page` [31 August 2026].

[2] City of New York. n.d. *Terms of Use*. Available at: `https://www.nyc.gov/main/terms-of-use` [Accessed 31 August 2026].

# A Appendix A: EDA Visualizations

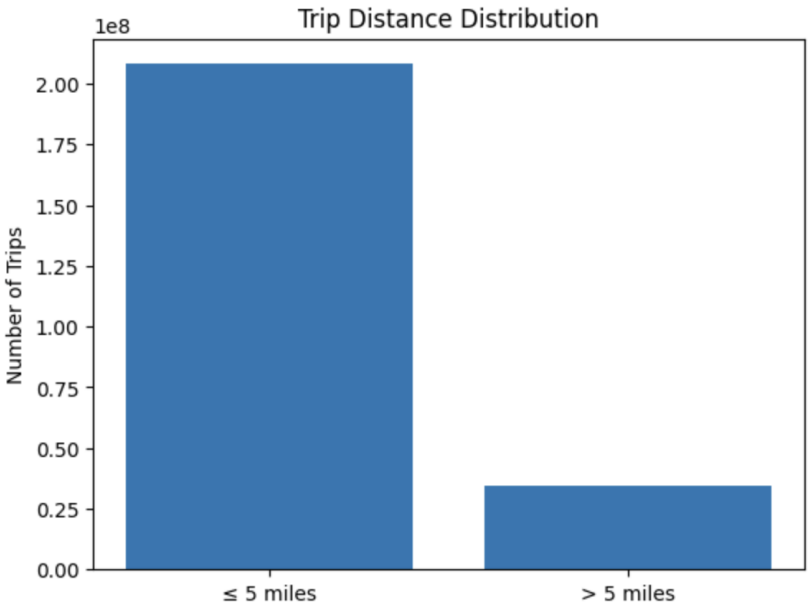



Figure 1: Distribution of Trip Distance

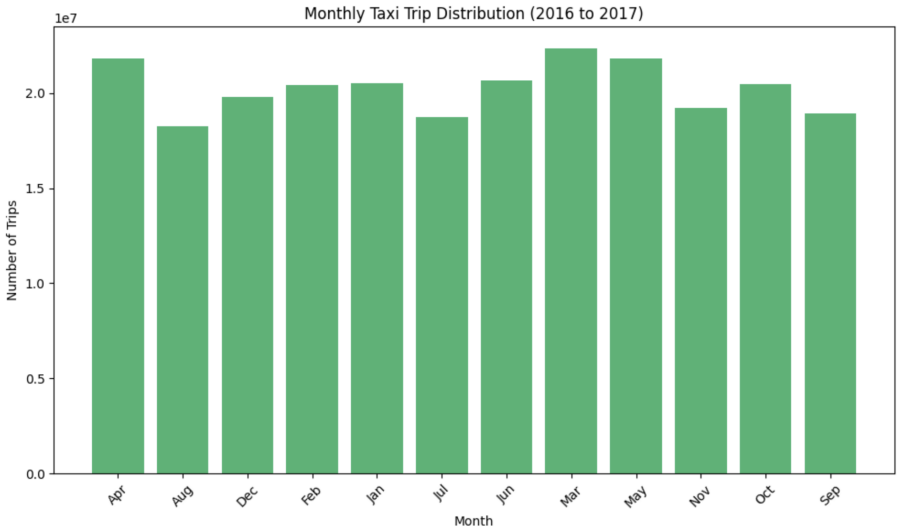



Figure 2: Monthly Trip Distribution

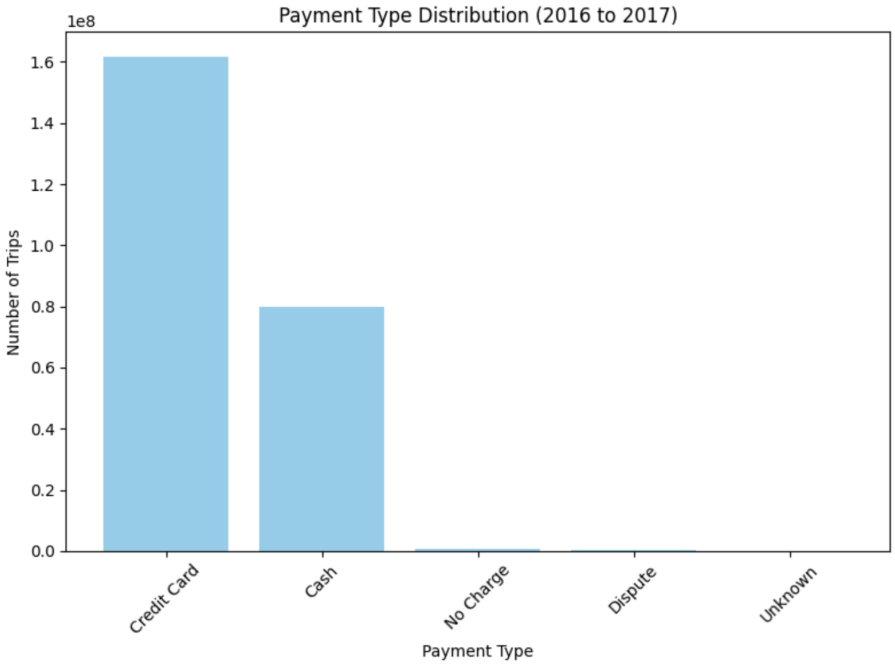



Figure 3: Payment Type Distribution

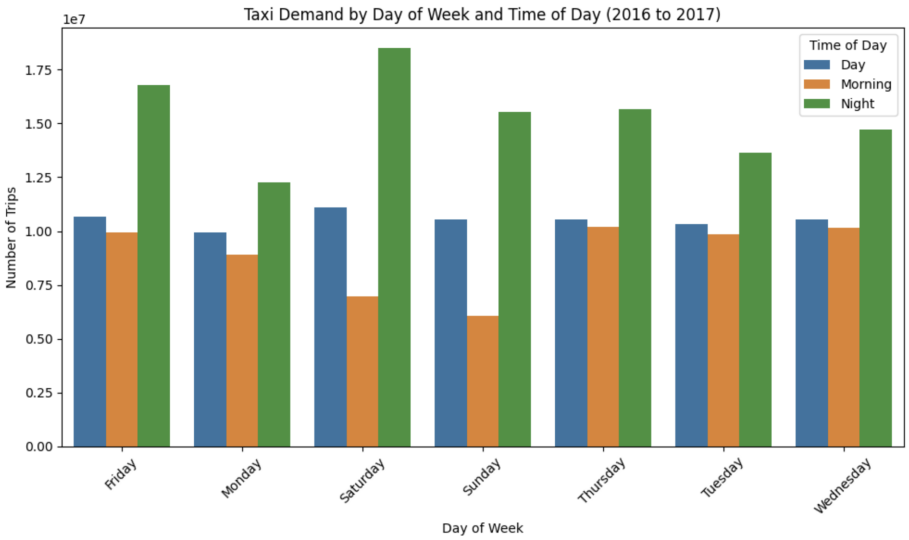



Figure 4: Weekly Taxi Demand